

# NON-MF GROUPS

SAUERS

We show that not every countable group is MF, where MF means admitting finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. The main tool is a one-sided compression criterion that places normal Kazhdan subgroups in the MF radical. Applied to the binary Leavitt group, it shows that the finitely generated simple property-(T) group  $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$  has trivial image under every homomorphism to an MF group; consequently its reduced group  $C^*$ -algebra is a separable stably finite  $C^*$ -algebra that is not MF. A second application gives a finitely presented torsion-free group with property (T) that has trivial image under every homomorphism to an MF group, and every nontrivial quotient of it has the same property.

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CONSUMPTION**

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## 1. INTRODUCTION

Not every countable group is MF. The counterexample is

$$H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2)).$$

The group  $H$  is finitely generated, nontrivial, simple, and has property (T), but every homomorphism from  $H$  to an MF group is trivial. In particular,  $H$  is not MF. Its reduced group  $C^*$ -algebra  $C_r^*(H)$  is separable and stably finite, but is not MF. The group is not new: by the isomorphisms of Khanh–Thanh,  $H$  is the unit group of the binary Leavitt algebra [K–T], with which OpenAI constructed the first nonsofic group [OAI]. The theorems about it are.

More precisely, a countable group  $G$  is *MF* if there are positive integers  $d_n$  and maps  $V_n: G \rightarrow \mathcal{U}(d_n)$ , with  $V_n(1) = 1$ , such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \quad (g \in G \setminus \{1\}).$$

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This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE]. Here, MF group always means this operator-norm local-approximation property. The strong-convergence convention also requires the matrix models to recover the norms of the left regular representation [GKM, Sch].

Equivalently,  $G$  embeds in the unitary group of the norm matrix corona associated to the sequence  $\mathbf{d} = (d_n)$ ,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \rightarrow 0\}$$

is the  $c_0$ -direct sum.

A separable  $C^*$ -algebra is called MF if it embeds as a  $C^*$ -subalgebra of a norm matrix corona. For every countable group  $G$ , the reduced group algebra  $C_r^*(G)$  is separable and has a faithful canonical trace, hence is stably finite. A non-MF group therefore gives a separable stably finite reduced group  $C^*$ -algebra that is not MF (see the proof of Theorem B).

For a countable group  $G$ , define its MF radical by

$$\text{Rad}_{\text{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

We call a homomorphism  $G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$  a *corona homomorphism*. Proposition 2.1 shows that  $G$  is MF precisely when  $\text{Rad}_{\text{MF}}(G)$  is trivial. If  $\text{Rad}_{\text{MF}}(G) = G$ , every homomorphism from  $G$  to an MF group is trivial.

For a subgroup  $L \leq G$ , put

$$\text{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression-centralizer defect* by

$$(1) \quad \mathfrak{D}_G(L) = \langle\langle [ucu^{-1}, \ell] : u \in \text{Comp}_G(L), c \in C_G(L), \ell \in L \rangle\rangle_G.$$

**Theorem A** (one-sided compression criterion). *Let  $G$  be countable and let  $L \leq G$  have property (T). If  $K \trianglelefteq G$  has property (T) and  $K \leq \mathfrak{D}_G(L)$ , then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

*In particular, a nontrivial such  $K$  makes  $G$  non-MF. If  $G$  has property (T) and  $\mathfrak{D}_G(L) = G$ , then  $\text{Rad}_{\text{MF}}(G) = G$ .*

We sketch the argument. Corollary 2.6 gives  $\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G)$ . That is, every operator norm asymptotic representation of  $G$  is asymptotically trivial on  $\mathfrak{D}_G(L)$  in normalized Hilbert–Schmidt norm. This holds because conjugation by the compressor moves the Kazhdan projection of  $L$  below itself, and stable finiteness of the corona forces equality (Section 2.3). If a corona homomorphism were nontrivial on  $K$ , the Kazhdan projection of  $K$  would cut out a  $G$ -invariant corner on which  $K$  has no nonzero fixed vectors (Lemma 2.7); by the Kazhdan inequality some  $s_0 \in K$  would then

stay bounded away from the identity in normalized Hilbert–Schmidt norm along a subsequence, contradicting  $s_0 \in \mathfrak{D}_G(L)$ .

Theorem 2.2 gives a separate finite-dimensional vanishing result: every finite-dimensional linear representation of  $G$  over any field kills  $\mathfrak{D}_G(L)$ .

Property (T) is not itself an obstruction to MF approximation: every residually finite group is MF, including many Kazhdan groups. In the argument above, property (T) controls the fixed vectors of an exact corona representation. The one-sided compression forces the corresponding Kazhdan projection to commute with the compressor, which makes the Hilbert–Schmidt commutant invariant.

We next apply Theorem A to the group  $H$ .

Let  $L_{\mathbb{F}_2}(1, 2)$  denote the binary Leavitt algebra. It is the unital  $\mathbb{F}_2$ -algebra generated by  $s_0, s_1, t_0, t_1$  subject to

$$(2) \quad t_i s_j = \delta_{ij}, \quad s_0 t_0 + s_1 t_1 = 1.$$

For a unital ring  $R$ , write  $e_{ij}(a) = 1 + aE_{ij}$  and let  $\text{EL}_n(R)$  be the subgroup of  $\text{GL}_n(R)$  generated by the elements  $e_{ij}(a)$ ,  $i \neq j$ .

**Theorem B** (the binary Leavitt group). *Put  $R = L_{\mathbb{F}_2}(1, 2)$  and  $H = \text{EL}_{12}(R)$ . Then  $H$  is finitely generated, nontrivial, simple, has property (T), and*

$$\text{Rad}_{\text{MF}}(H) = H.$$

*Equivalently, every homomorphism from  $H$  to an MF group is trivial. In particular,  $H$  is non-MF. Its reduced group  $C^*$ -algebra  $C_r^*(H)$  is separable and stably finite, but is not MF.*

Inside  $H$ , an explicit element  $\tau$  satisfies  $\tau L \tau^{-1} \leq L$  for the subgroup  $L \cong \text{EL}_3(R)$  in the upper left. The element  $c = e_{34}(1)$  centralizes  $L$ , and for  $\ell = e_{12}(1)$  the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The relation  $t_1(s_1 t_1)s_1 = 1$  and the Steinberg commutator relations show directly that  $d$  normally generates  $H$ . Since  $d \in \mathfrak{D}_H(L)$  and  $\mathfrak{D}_H(L)$  is normal, it follows that  $\mathfrak{D}_H(L) = H$ . Therefore Theorem A gives  $\text{Rad}_{\text{MF}}(H) = H$ .

**Theorem C** (a torsion-free group with full MF radical). *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group  $Q$  with property (T) and*

$$\text{Rad}_{\text{MF}}(Q) = Q.$$

*Every nontrivial quotient of  $Q$  also equals its own MF radical; in particular, no nontrivial quotient of  $Q$  is MF.*

The group  $Q$  is a small-cancellation quotient, in the sense of Hull [Hu], of the torsion-free property-(T) group of Fournier-Facio [FF, §2], chosen so that the compression defect normally generates the quotient (Section 4). Thus Theorem A also applies to  $Q$ .

**Relation to prior work.** Blackadar and Kirchberg developed the notions of MF and NF algebras and proved that a separable  $C^*$ -algebra is NF if and only if it is nuclear and MF [BK]. It also asked whether every separable nuclear stably finite  $C^*$ -algebra is quasidiagonal. This was a question about nuclear  $C^*$ -algebras, not about arbitrary groups.

The group property used here was introduced by Carrión–Dadarlat–Eckhardt [CDE]. If every separable stably finite  $C^*$ -algebra were MF, then every countable group would be MF because its reduced group  $C^*$ -algebra is separable and stably finite. Later papers on approximation of groups relate the resulting operator norm problem to Kirchberg’s question [LO, Th]. The negative solution of the Connes embedding problem [JNVWY] yields separable stably finite  $C^*$ -algebras that are not MF, as Goldbring and Hart observe [GH, Proposition 6.1 and Remark 6.2]. Theorem B gives such an example among reduced group  $C^*$ -algebras. Korchagin subsequently developed permanence and local approximation results for MF groups [Kor]. Shulman proves that  $A *_C A$  is MF whenever  $A$  is a separable MF  $C^*$ -algebra and  $C \subseteq A$  is a  $C^*$ -subalgebra [Sh, Theorem 10]. She also proves that the full group  $C^*$ -algebras of amalgamated free products of amenable groups and of certain semidirect products are MF [Sh]. Nonapproximability was known for the unnormalized Schatten  $p$ -norms with  $1 < p < \infty$  [DGLT, LO]. Bachner–Dogon–Lubotzky later proved nonapproximability for  $p = 1$  [BDL]. The operator norm case  $p = \infty$  remained open. They also identify stability from operator norm to normalized Hilbert–Schmidt norm as a possible route to a non-MF group [BDL, Proposition 1.5]; that is the route taken here.

Leavitt introduced the algebras  $L_K(1, n)$  as examples of rings with nonunique module rank [Lev]. The compression used here comes from their defining relations. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser classified the subgroups of  $GL_n$  over an exchange ring that are normalized by  $EL_n$  [Pre]. His sandwich theorem, together with the simplicity of the binary Leavitt algebra and the equality  $Z(R) = \mathbb{F}_2$ , implies that  $EL_{12}(R)$  is simple (Proposition 3.3). Ershov–Jaikin–Zapirain give property (T) for  $EL_n(R)$  whenever  $n \geq 3$  and  $R$  is a finitely generated unital associative ring [EJZ, Theorem 1.1]. OpenAI used the binary Leavitt unit group to construct the first known nonsofic group [OAI, Chapter 3]. Khanh–Thanh’s matrix-ring isomorphisms and their equality  $GL_n(R) = EL_n(R)$  identify that group with the group  $H$  studied here [K–T, Proposition 4.2 and Corollary 4.4]. We apply the Leavitt compression to operator norm approximation. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

## 2. ONE-SIDED COMPRESSION AND THE MF RADICAL

**2.1. The MF radical.** The image of a corona homomorphism from  $G$  is a countable subgroup of a corona unitary group, hence MF; conversely, every

countable MF group embeds in the unitary group of a norm matrix corona. Thus  $\text{Rad}_{\text{MF}}(G)$  is also the intersection of the kernels of all homomorphisms from  $G$  to MF groups.

**Proposition 2.1** (basic properties of the MF radical). *For every countable group  $G$ , the subgroup  $\text{Rad}_{\text{MF}}(G)$  is fully invariant, the quotient  $G/\text{Rad}_{\text{MF}}(G)$  is MF, and  $G$  is MF if and only if  $\text{Rad}_{\text{MF}}(G)$  is trivial.*

*Proof.* An intersection of kernels is normal. If  $\alpha$  is an endomorphism of  $G$ ,  $x \in \text{Rad}_{\text{MF}}(G)$ , and  $\pi$  is a corona homomorphism from  $G$ , then  $\pi \circ \alpha$  kills  $x$ . Hence  $\pi(\alpha(x)) = 1$  for every  $\pi$ , which proves full invariance.

Put  $R = \text{Rad}_{\text{MF}}(G)$ . The trivial group is MF. If  $G/R$  is nontrivial, let  $x \in G/R$  be nontrivial and choose  $g \in G$  mapping to  $x$ . Then  $g \notin R$ . Some homomorphism from  $G$  to an MF group therefore separates  $g$  from the identity. It kills  $R$  and descends to a homomorphism from  $G/R$  that separates  $x$ . Thus  $G/R$  is residually MF, and hence MF by Korchagin [Kor, Proposition 6].

If  $G$  is MF, a corona embedding of  $G$  has trivial kernel, so  $\text{Rad}_{\text{MF}}(G) = 1$ . Conversely, if  $\text{Rad}_{\text{MF}}(G) = 1$ , then  $G = G/\text{Rad}_{\text{MF}}(G)$  is MF.  $\square$

## 2.2. Finite dimension.

**Theorem 2.2** (finite-dimensional commutant rigidity). *Let  $G$  be a group, let  $L \leq G$ , and suppose  $u \in G$  satisfies  $uLu^{-1} \leq L$ . If  $k$  is a field,  $V$  is a finite-dimensional  $k$ -vector space, and  $\rho: G \rightarrow \text{GL}(V)$  is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where  $\rho(L)'$  is the commutant of  $\rho(L)$  in  $\text{End}_k(V)$ . Consequently, if  $c \in C_G(L)$ , then

$$\rho([ucu^{-1}, \ell]) = 1 \quad (\ell \in L).$$

*Proof.* Put  $C = \rho(L)'$ . If  $x \in C$ ,  $h \in L$ , and  $h' = uhu^{-1} \in L$ , then

$$\begin{aligned} \rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u) \\ &= \rho(u)^{-1}x\rho(h')\rho(u) \\ &= \rho(u)^{-1}x\rho(u)\rho(h). \end{aligned}$$

Thus  $\rho(u)^{-1}C\rho(u) \subseteq C$ . Conjugation by  $\rho(u)^{-1}$  is injective, and  $C$  is finite-dimensional. Its restriction to  $C$  is therefore surjective, so the inclusion is equality. Conjugating by  $\rho(u)$  gives  $\rho(u)C\rho(u)^{-1} = C$ .

If  $c \in C_G(L)$ , then  $\rho(c) \in C$ , hence  $\rho(ucu^{-1}) \in C$ . Therefore

$$\rho([ucu^{-1}, \ell]) = 1 \quad (\ell \in L).$$

$\square$

Finite dimensionality is essential: an injective linear self-map of an infinite-dimensional commutant need not be surjective.

**2.3. Kazhdan transport in normalized Hilbert–Schmidt norm.** For  $a \in M_d(\mathbb{C})$  write

$$\|a\|_2 = \text{tr}_d(a^*a)^{1/2}, \quad \text{tr}_d = \frac{1}{d} \text{Tr}.$$

An *operator norm asymptotic representation* of a countable group  $G$  is a sequence of maps  $V_n: G \rightarrow \mathcal{U}(d_n)$  such that  $V_n(1) = 1$  and

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \rightarrow 0\}.$$

The operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, so  $K_2(V)$  is a normal subgroup of  $G$ . Define

$$(3) \quad R_{\infty \rightarrow 2}(G) = \bigcap_V K_2(V),$$

where  $V$  ranges over all operator norm asymptotic representations of  $G$ .

For  $L \leq G$ , let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \left| \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \rightarrow 0 \quad (\ell \in L) \end{array} \right. \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of  $V(L)$ . Coordinatewise conjugation by  $V_n(g)$  defines a bijection, denoted  $\text{Ad}(V(g))$ , of the bounded matrix sequences.

**Lemma 2.3.** *For every sequence of positive integers  $(m_n)$ , the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

*is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.*

*Proof.* Suppose that  $v^*v = 1$  in the corona, and let  $(x_n)$  be a bounded lift of  $v$ . Then  $x_n^*x_n \rightarrow 1$  in norm. For all sufficiently large  $n$ , the matrix  $x_n$  is invertible, and the unitary  $x_n(x_n^*x_n)^{-1/2}$  differs from  $x_n$  by  $o(1)$ . Thus  $v$  is represented by unitaries and  $vv^* = 1$ . The same argument applies to every matrix amplification of the corona. If  $A$  is a stably finite unital  $C^*$ -algebra, equivalent projections  $p \leq q$  satisfy  $p = q$ . Indeed, if  $w^*w = q$  and  $ww^* = p$ , then  $w \in qAq$  is an isometry in the corner. The corner is finite because  $w + (1 - q)$  is an isometry in  $A$ . Consequently  $ww^* = q$ , and hence  $p = q$ .  $\square$

**Lemma 2.4** (one-sided order for the Kazhdan projection). *Let  $L$  have property (T), let  $B$  be a unital  $C^*$ -algebra, and let  $\pi: L \rightarrow \mathcal{U}(B)$  be a homomorphism. Denote by  $P \in B$  the image of the Kazhdan projection under the extension  $C_{\max}^*(L) \rightarrow B$ . If  $U \in \mathcal{U}(B)$  satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \leq P.$$

*Proof.* Represent  $B$  faithfully and nondegenerately on a Hilbert space  $\mathcal{H}$ . The represented projection  $P$  is the orthogonal projection onto  $\text{Fix } \pi(L)$ . If  $\xi$  is  $L$ -fixed and  $\ell \in L$ , then

$$\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi,$$

because  $U\pi(\ell)U^*$  belongs to  $\pi(L)$ . Hence  $U^*\text{Fix } \pi(L) \subseteq \text{Fix } \pi(L)$ , so the range projection  $U^*PU$  is dominated by  $P$  in  $B(\mathcal{H})$ . Faithfulness of the representation gives the same projection inequality in  $B$ .  $\square$

**Theorem 2.5** (one-sided Kazhdan transport). *Let  $L \leq G$  have property (T), and let  $u \in G$  satisfy  $uLu^{-1} \leq L$ . Let  $(V_n)$  be an operator norm asymptotic representation of  $G$ . Then*

$$\text{Ad}(V(u))(\mathcal{C}_2(V, L)) = \mathcal{C}_2(V, L).$$

*Proof.* Fix  $x = (x_n) \in \mathcal{C}_2(V, L)$  and let  $\omega$  be a free ultrafilter. Regard  $M_{d_n}(\mathbb{C})$  as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces  $\mathcal{K}_\omega$ . Conjugation by the  $V_n(g)$  defines a unitary representation  $\sigma$  of  $G$  on  $\mathcal{K}_\omega$ . Indeed, for unitaries  $A, B$  one has

$$\|\text{Ad}(A) - \text{Ad}(B)\| \leq 2\|A - B\|,$$

so the conjugation actions are multiplicative modulo  $c_0$  because the multiplicative defects of  $(V_n)$  tend to zero in operator norm. The class  $\xi = [x_n]_\omega$  is fixed by  $\sigma(L)$ .

For  $g \in G$ , let  $\tilde{\sigma}(g)$  be the class of the coordinate operators  $\text{Ad}(V_n(g))$  in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each  $M_{d_n}(\mathbb{C})$ , this is a norm matrix corona with coordinate sizes  $d_n^2$ . By the preceding estimate,  $g \mapsto \tilde{\sigma}(g)$  is a homomorphism  $G \rightarrow \mathcal{U}(\mathcal{B})$ . Its restriction to  $L$  therefore extends to a  $*$ -homomorphism  $C_{\max}^*(L) \rightarrow \mathcal{B}$ . Let  $P \in \mathcal{B}$  be the image of the Kazhdan projection [AW]. Under the natural representation of  $\mathcal{B}$  on  $\mathcal{K}_\omega$ , its range is  $\text{Fix } \sigma(L)$ . The projection identity below is proved inside  $\mathcal{B}$ ; the representation of  $\mathcal{B}$  on  $\mathcal{K}_\omega$  is used only afterwards. Since  $uLu^{-1} \leq L$ , the unitary  $U = \tilde{\sigma}(u)$  satisfies the hypothesis of Lemma 2.4, and hence  $U^*PU \leq P$  inside  $\mathcal{B}$ . The two projections are unitarily equivalent. Lemma 2.3 therefore gives  $U^*PU = P$ , so  $U$  commutes with  $P$ . Hence both  $U\xi$  and  $U^*\xi$  are fixed by  $\sigma(L)$ . Since  $\omega$  was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence. Thus  $\text{Ad}(V(u))$  and its inverse both preserve  $\mathcal{C}_2(V, L)$ , which proves the equality.  $\square$

**Corollary 2.6.** *Let  $L \leq G$  have property (T), let  $uLu^{-1} \leq L$ , and let  $c \in C_G(L)$ . Then*

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G) \quad (\ell \in L).$$

*Proof.* Let  $(V_n)$  be an operator norm asymptotic representation. If  $c \in C_G(L)$ , then  $(V_n(c)) \in \mathcal{C}_2(V, L)$ . By Theorem 2.5, the same is true of  $(V_n(u)V_n(c)V_n(u)^*)$ . Asymptotic multiplicativity identifies this sequence in operator norm with  $(V_n(ucu^{-1}))$ ; explicitly,

$$\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\| \longrightarrow 0.$$

Therefore

$$\|V_n([ucu^{-1}, \ell]) - 1\|_2 \longrightarrow 0 \quad (\ell \in L).$$

Intersecting over  $(V_n)$  proves the claim.  $\square$

**2.4. Normal Kazhdan subgroups and the MF radical.** Let  $K \trianglelefteq G$  have property (T), and let  $p$  be the image of its Kazhdan projection under a corona homomorphism of  $G$ . Normality of  $K$  implies that  $p$  commutes with the image of  $G$ . Hence  $q = 1 - p$  also commutes with that image, and in every nondegenerate representation of the corner  $q\mathcal{Q}_{\mathbf{d}}q$  the induced representation of  $K$  has no nonzero fixed vectors.

**Lemma 2.7** (central corona corners). *Let  $\rho: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$  be a homomorphism from a countable group, and let  $q \in \mathcal{Q}_{\mathbf{d}}$  be a nonzero projection commuting with  $\rho(G)$ . Then, after passing to an infinite coordinate subsequence, there are nonzero projections  $q_n \in M_{d_n}(\mathbb{C})$ , integers  $r_n = \text{rank}(q_n)$ , unitary identifications  $J_n: \mathbb{C}^{r_n} \rightarrow q_n\mathbb{C}^{d_n}$ , and an operator norm asymptotic representation*

$$W_n: G \longrightarrow \mathcal{U}(r_n)$$

*with  $W_n(1) = I_{r_n}$ . In the corona over the retained coordinates, the class of  $(J_n W_n(g) J_n^*)$  is the coordinate restriction of  $q\rho(g)$ .*

*Proof.* Lift  $q$  to projections  $q_n$ , and lift each  $\rho(g)$  to unitaries  $U_n(g)$ , choosing  $U_n(1) = I_{d_n}$ . The first lift is obtained by spectral rounding; the second is obtained by polar correction, since the two unitarity defects of any lift converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \longrightarrow 0 \quad (g \in G).$$

Consequently  $q_n U_n(g) q_n$ , viewed on  $q_n \mathbb{C}^{d_n}$ , has both unitarity defects converging to zero. For each fixed  $g$ , let  $\widetilde{W}_n(g)$  be its polar correction whenever both defects are at most  $1/2$ , and set  $\widetilde{W}_n(g) = q_n$  at the finitely many earlier stages. Then  $\widetilde{W}_n(g)$  is unitary in the corner and satisfies

$$\|\widetilde{W}_n(g) - q_n U_n(g) q_n\| \longrightarrow 0.$$

For  $g = 1$ , the compression is already the identity  $q_n$  of the corner, so take  $\widetilde{W}_n(1) = q_n$ . Since  $q \neq 0$ , infinitely many  $q_n$  are nonzero. Retain and relabel



those coordinates. Put  $r_n = \text{rank}(q_n)$ , choose a unitary  $J_n: \mathbb{C}^{r_n} \rightarrow q_n \mathbb{C}^{d_n}$ , and define

$$W_n(g) = J_n^* \widetilde{W}_n(g) J_n \in \mathcal{U}(r_n).$$

Conjugation by  $J_n$  preserves operator norms and multiplicative defects. Thus  $(W_n)$  is an operator norm asymptotic representation with  $W_n(1) = I_{r_n}$ . The displayed estimate identifies the class of  $(J_n W_n(g) J_n^*)$  with the coordinate restriction of  $q\rho(g)$ .  $\square$

**Theorem 2.8** (normal Kazhdan radical theorem). *Let  $G$  be countable and let  $K \trianglelefteq G$  have property (T) with  $K \leq R_{\infty \rightarrow 2}(G)$ . Then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

*Proof.* Suppose that a homomorphism  $\Theta: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$  is nontrivial on  $K$ . Replace  $G$  by  $\Theta(G)$ , replace  $K$  by  $\Theta(K)$ , and denote the resulting inclusion into the corona again by  $\Theta$ . The new  $K$  is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. It is contained in  $R_{\infty \rightarrow 2}(\Theta(G))$ , because an operator norm asymptotic representation of  $\Theta(G)$ , precomposed with the quotient map, is one of the original group.

Let  $p_K \in C_{\max}^*(K)$  be the Kazhdan projection and let  $p$  be its image in  $\mathcal{Q}_{\mathbf{d}}$ . Under any faithful representation of the corona,  $p$  is the projection onto the  $K$ -fixed vectors. Normality of  $K$  gives

$$\Theta(g)p\Theta(g)^* = p \quad (g \in G).$$

Indeed, in a faithful representation of the corona the two projections have ranges  $\Theta(g) \text{Fix } \Theta(K)$  and  $\text{Fix } \Theta(K)$ ; these agree because  $gKg^{-1} = K$ . Put  $q = 1 - p$ . The projection  $q$  is nonzero: if  $p = 1$ , then every vector is fixed by  $K$ , so every element of  $K$  has trivial corona image. Lemma 2.7 gives positive integers  $r_n$  and an operator norm asymptotic representation  $W_n: G \rightarrow \mathcal{U}(r_n)$ . Let  $\widehat{\Theta}: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{r}})$  be its corona homomorphism. Under the coordinatewise corner identifications,  $\widehat{\Theta}$  is the coordinate restriction of  $g \mapsto q\Theta(g)$ .

Choose a finite Kazhdan set  $S \subseteq K$  and a Kazhdan constant  $\kappa > 0$ . Represent  $\mathcal{Q}_{\mathbf{r}}$  faithfully. The Kazhdan projection of  $K$  under  $\widehat{\Theta}$  is zero, so  $\widehat{\Theta}|_K$  has no nonzero fixed vectors, and the Kazhdan inequality gives, for the positive element

$$b = \frac{1}{|S|} \sum_{s \in S} (\widehat{\Theta}(s) - 1)^* (\widehat{\Theta}(s) - 1)$$

the inequality

$$b \geq \frac{\kappa^2}{|S|} 1.$$

The coordinate elements

$$b_n = \frac{1}{|S|} \sum_{s \in S} (W_n(s) - I_{r_n})^* (W_n(s) - I_{r_n})$$

represent  $b$ . Taking the normalized trace on  $M_{r_n}(\mathbb{C})$  yields

$$\frac{1}{|S|} \sum_{s \in S} \|W_n(s) - I_{r_n}\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

The order inequality for  $b$  says that the negative part of  $b_n - (\kappa^2/|S|)I_{r_n}$  converges to zero in operator norm. Taking normalized traces gives the displayed inequality. After passing to a subsequence, one fixed  $s_0 \in S$  therefore stays a positive Hilbert–Schmidt distance from  $I_{r_n}$ . This contradicts  $s_0 \in K \leq R_{\infty \rightarrow 2}(G)$ , established after passing to  $\Theta(G)$ . Thus every corona homomorphism kills  $K$ , as required.  $\square$

*Proof of Theorem A.* For every  $u \in \text{Comp}_G(L)$ ,  $c \in C_G(L)$ , and  $\ell \in L$ , Corollary 2.6 gives

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G).$$

Normality of  $R_{\infty \rightarrow 2}(G)$  therefore gives

$$\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G).$$

Since  $K \leq \mathfrak{D}_G(L)$ , Theorem 2.8 now gives  $K \leq \text{Rad}_{\text{MF}}(G)$ . Taking a nontrivial  $K$  gives the non-MF conclusion. Taking  $K = G = \mathfrak{D}_G(L)$  gives  $\text{Rad}_{\text{MF}}(G) = G$ .  $\square$

By Theorem 2.2, every finite-dimensional linear representation of  $G$  kills the generators of  $\mathfrak{D}_G(L)$ , and its kernel is normal, so it kills  $\mathfrak{D}_G(L)$ . Consequently,  $\mathfrak{D}_G(L) = 1$  if  $G$  has a faithful finite-dimensional linear representation. The same conclusion holds if  $G$  is residually finite, because every nonidentity element then survives in a finite permutation representation. If  $G$  is amenable, its property-(T) subgroup  $L$  is finite [BHV]; then  $uLu^{-1} = L$ , so  $ucu^{-1}$  centralizes  $L$  for every  $u \in \text{Comp}_G(L)$ , and again  $\mathfrak{D}_G(L) = 1$ .

The argument is specific to operator norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt approximations do not provide the operator norm control used to construct the conjugation representation in Theorem 2.5.

### 3. THE BINARY LEAVITT GROUP

The relation  $s_0t_0 + s_1t_1 = 1$  of the binary Leavitt algebra gives an element  $\tau \in H$  with  $\tau L \tau^{-1} \leq L$  for a copy  $L$  of  $\text{EL}_3(R)$ , and a commutator  $d \neq 1$  in  $\mathfrak{D}_H(L)$ ; since  $d$  normally generates  $H$ ,  $\mathfrak{D}_H(L) = H$ , and Theorem A applies.

**3.1. The self-compression.** Throughout this section  $R = L_{\mathbb{F}_2}(1, 2)$  and

$$p = s_0t_0, \quad q = s_1t_1.$$

The relations (2) give

$$(4) \quad p + q = 1, \quad t_1qs_1 = 1.$$

In particular  $q \neq 0$ .

We use indices  $0, \dots, 11$  for  $12 \times 12$  matrices. We identify  $\mathrm{GL}_3(R)$  with the subgroup of  $\mathrm{GL}_{12}(R)$  consisting of the matrices  $\mathrm{diag}(A, I_9)$ . Define a map  $\Psi: M_3(R) \rightarrow M_3(R)$  by

$$(5) \quad \Psi(A) = qI_3 + s_0At_0.$$

Here  $s_0At_0$  denotes the matrix  $(s_0a_{ij}t_0)_{ij}$ . The relations  $q^2 = q$ ,  $qs_0 = t_0q = 0$ , and  $t_0s_0 = 1$ , all consequences of (2), show that  $\Psi$  is multiplicative. Moreover,

$$\Psi(I_3) = qI_3 + s_0I_3t_0 = qI_3 + pI_3 = (p+q)I_3 = I_3,$$

so  $\Psi$  preserves the identity. Finally,  $t_0\Psi(A)s_0 = A$ , so  $\Psi$  is injective. Hence  $\Psi$  restricts to an injective group homomorphism  $\mathrm{GL}_3(R) \rightarrow \mathrm{GL}_3(R)$ .

To realize  $\Psi$  by conjugation on the embedded copy of  $\mathrm{GL}_3(R)$ , define the following  $6 \times 6$  block matrices:

$$X = \begin{pmatrix} s_0I_3 & s_1t_0I_3 \\ 0 & t_1I_3 \end{pmatrix}, \quad Y = \begin{pmatrix} t_0I_3 & 0 \\ s_0t_1I_3 & s_1I_3 \end{pmatrix}.$$

A block multiplication using (2) gives  $Y = X^{-1}$ . Put

$$(6) \quad \tau = \mathrm{diag}(X, Y) \in \mathrm{GL}_{12}(R).$$

A direct block multiplication gives

$$(7) \quad \tau \mathrm{diag}(A, I_9) \tau^{-1} = \mathrm{diag}(\Psi(A), I_9).$$

**Lemma 3.1.** *The matrix  $\tau$  defined in (6) belongs to  $\mathrm{EL}_{12}(R)$ .*

*Proof.* For  $B \in M_6(R)$ , the block matrices  $\begin{pmatrix} I & B \\ 0 & I \end{pmatrix}$  and  $\begin{pmatrix} I & 0 \\ B & I \end{pmatrix}$  lie in  $\mathrm{EL}_{12}(R)$ : they are the products of the elementary matrices  $e_{r,6+s}(b_{rs})$ , respectively  $e_{6+r,s}(b_{rs})$ , and no cross terms arise because distinct off-diagonal matrix units have product zero. Whitehead's identity

$$\mathrm{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}$$

with  $I = I_6$  therefore expresses  $\tau$  as a product of six matrices of  $\mathrm{EL}_{12}(R)$ .  $\square$

Set  $H = \mathrm{EL}_{12}(R)$ , and let  $L = \mathrm{EL}_3(R) \leq H$  denote the image of the embedding  $A \mapsto \mathrm{diag}(A, I_9)$ .

**Proposition 3.2.** *The groups  $H$  and  $L$  have property (T), and  $\tau \in H$  satisfies*

$$(8) \quad \tau L \tau^{-1} \leq L.$$

*Proof.* The ring  $R$  is generated by  $s_0, s_1, t_0, t_1$  as a unital ring. The theorem of Ershov and Jaikin-Zapirain therefore gives property (T) for  $\mathrm{EL}_{12}(R)$  and  $\mathrm{EL}_3(R)$  [EJZ, Theorem 1.1]. Lemma 3.1 gives  $\tau \in H$ .

For  $i \neq j$  and  $a \in R$ , equations (7) and (5) give

$$\tau e_{ij}(a) \tau^{-1} = e_{ij}(s_0at_0) \in L,$$

where the elementary matrices are viewed in the embedded copy of  $L$ . These matrices generate  $L$ , so (8) follows.  $\square$

### 3.2. Simplicity and the full MF radical.

**Proposition 3.3.** *The group  $H = \text{EL}_{12}(R)$  is nontrivial and simple.*

*Proof.* The ring  $R$  is purely infinite simple [AA], and therefore is an exchange ring [Ara]. Let  $N \trianglelefteq H$ . Since  $H = \text{EL}_{12}(R)$ , the subgroup  $N \leq \text{GL}_{12}(R)$  is normalized by  $\text{EL}_{12}(R)$ . Preusser's sandwich theorem [Pre, Theorem 3] provides an ideal  $I \trianglelefteq R$  such that

$$\text{EL}_{12}(R, I) \leq N \leq C_{12}(R, I).$$

Simplicity of  $R$  gives  $I = 0$  or  $I = R$ . If  $I = R$ , then  $\text{EL}_{12}(R, I) = H$ , so  $N = H$ .

Suppose that  $I = 0$ . Then  $N \leq C_{12}(R, 0) = Z(\text{GL}_{12}(R))$ . A matrix that commutes with every  $e_{ij}(1)$  is a scalar matrix  $\lambda I_{12}$ . Commutation with  $e_{ij}(a)$  for arbitrary  $a \in R$  then gives  $\lambda a = a\lambda$ , so  $\lambda \in Z(R)^\times$ . Since  $Z(R) = \mathbb{F}_2$  [AC, Corollary 4.3], we have  $\lambda = 1$  and  $N = 1$ . Finally,  $e_{01}(1) \neq 1$ , so  $H$  is nontrivial.  $\square$

**Proposition 3.4.** *Let*

$$c = e_{34}(1), \quad \ell = e_{12}(1), \quad d = [\tau c \tau^{-1}, \ell]$$

*in  $H$ . Then*

$$c \in C_H(L), \quad d = e_{02}(q) \neq 1, \quad \langle\langle d \rangle\rangle_H = H.$$

*Proof.* For  $i, j \in \{0, 1, 2\}$  with  $i \neq j$  and  $a \in R$ , we have  $j \neq 3$  and  $i \neq 4$ . Hence the Steinberg relations give

$$[e_{ij}(a), e_{34}(1)] = 1.$$

These matrices generate  $L$ , so  $c = e_{34}(1)$  belongs to  $C_H(L)$ .

Direct multiplication of the matrices in (6) gives

$$(9) \quad \tau c \tau^{-1} = e_{01}(q) e_{34}(1).$$

The second factor commutes with  $\ell = e_{12}(1)$ . The identity  $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$  therefore gives  $d = e_{02}(q)$ , which is nontrivial because  $q \neq 0$ .

Let  $N = \langle\langle d \rangle\rangle_H$ . Since  $t_1 q s_1 = 1$ , the Steinberg relations give

$$[d, e_{23}(s_1)] = e_{03}(q s_1) \in N, \quad [e_{10}(t_1), e_{03}(q s_1)] = e_{13}(1) \in N.$$

Conjugation by elementary Weyl matrices gives  $e_{ij}(1) \in N$  for every  $i \neq j$ . For  $a \in R$  and distinct  $i, j, k$ ,

$$[e_{ik}(a), e_{kj}(1)] = e_{ij}(a) \in N.$$

Thus  $N$  contains every elementary generator of  $H$ , so  $N = H$ .  $\square$

*Proof of Theorem B.* Let  $Y$  be the set of elements  $e_{ij}(a)$  with  $i \neq j$  and  $a \in \{1, s_0, s_1, t_0, t_1\}$ . Since  $e_{ij}(a + b) = e_{ij}(a)e_{ij}(b)$  and  $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$  for distinct  $i, j, k$ , and since every element of  $R$  is a sum of monomials in  $s_0, s_1, t_0, t_1$ , induction on the length of a monomial shows that every  $e_{ij}(a)$  lies in  $\langle Y \rangle$ ; there is always a third index because  $12 \geq 3$ . Thus  $H = \langle Y \rangle$  is finitely

generated, and  $H$  has property (T) by the Ershov–Jaikin–Zapirain theorem for elementary groups over finitely generated rings [EJZ, Theorem 1.1].

The containment (8), the centrality of  $c$  relative to  $L$ , and Proposition 3.4 show that  $d \in \mathfrak{D}_H(L)$ . The subgroup  $\mathfrak{D}_H(L)$  is normal and Proposition 3.4 gives  $\langle\langle d \rangle\rangle_H = H$ , so  $\mathfrak{D}_H(L) = H$ . The group  $H$  has property (T), so Theorem A, applied with  $K = H$ , gives  $\text{Rad}_{\text{MF}}(H) = H$ . Proposition 3.3 shows that  $H$  is nontrivial and simple.

If  $M$  is MF and  $f: H \rightarrow M$  is a homomorphism, compose  $f$  with a faithful corona embedding of  $M$ . Since  $\text{Rad}_{\text{MF}}(H) = H$ , this composition and hence  $f$  are trivial. If  $H$  were MF, its identity homomorphism would contradict this conclusion.

Finally,  $C_r^*(H)$  is separable because  $H$  is countable. Its canonical trace is faithful, so it is stably finite. If an MF embedding  $e: C_r^*(H) \rightarrow \mathcal{Q}_{\mathbf{d}}$  existed and  $p = e(1)$ , then  $h \mapsto e(\lambda_h) + (1 - p)$  would embed  $H$  in  $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ . This contradicts the fact that  $H$  is not MF.  $\square$

#### 4. A TORSION-FREE GROUP WITH FULL MF RADICAL

A group is acylindrically hyperbolic if it admits a non-elementary acylindrical action on a hyperbolic space [Os]. For such a group  $G$ , Hull [Hu, Theorem 3.12] provides a generating set  $A$  such that the Cayley graph  $\Gamma(G, A)$  is hyperbolic and the action of  $G$  on it is acylindrical and non-elementary; a subgroup  $N \leq G$  is *suitable* with respect to  $A$  if  $N$  acts non-elementarily on  $\Gamma(G, A)$  and normalizes no nontrivial finite subgroup of  $G$  [Hu, Definition 1.4]. We use Hull’s small cancellation theorem [Hu, Theorem 7.1] in the following form.

**Theorem 4.1** (Hull’s small cancellation theorem). *Let  $G$  be acylindrically hyperbolic, let  $N \leq G$  be suitable with respect to  $A$ , and let  $t_1, \dots, t_m \in G$  and a finite subset  $F \subseteq G$  be given. Then there is a surjective homomorphism  $q: G \rightarrow Q$  such that  $Q$  is acylindrically hyperbolic,  $q|_F$  is injective,  $q(t_i) \in q(N)$  for all  $i$ , and every element of finite order in  $Q$  is the image of an element of the same order in  $G$ .*

Moreover the kernel of  $q$  is the normal closure of  $m$  elements; this is read off from Hull’s proof: the case  $m = 1$  adds the single relator  $W = t_1^{-1}h_1^{m_1}h_2^{l_1} \cdots h_1^{m_n}h_2^{l_n}$  for suitable  $h_1, h_2 \in N$ , and the general case is an induction on  $m$ .

**Lemma 4.2** (saturation). *Let  $G$  be finitely presented, torsion-free, and acylindrically hyperbolic, let  $N \trianglelefteq G$  be nontrivial, and let  $F \subseteq G$  be finite. Then there is a surjective homomorphism  $q: G \rightarrow Q$  such that  $Q$  is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic,  $q|_F$  is injective, and  $q(N) = Q$ .*

*Proof.* Fix  $A$  as above. Since  $G$  is torsion-free and  $N \neq 1$ ,  $N$  is infinite, so  $|N \cap gNg^{-1}| = \infty$  for every  $g \in G$ , and by Osin [Os, Lemma 7.1]  $N$  acts non-elementarily on  $\Gamma(G, A)$ . The only finite subgroup of  $G$  is trivial, so

$N$  is suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8],  $N$  contains two elements  $h_1, h_2$  such that  $N_0 = \langle h_1, h_2 \rangle$  is again suitable. Apply Theorem 4.1 to  $N_0$  with  $t_1, \dots, t_m$  a finite generating set of  $G$  and with the set  $F$ . Then

$$Q = \langle q(t_1), \dots, q(t_m) \rangle \leq q(N_0) \leq q(N) \leq Q,$$

so  $Q = q(N_0) = \langle q(h_1), q(h_2) \rangle$  and  $q(N) = Q$ . The group  $Q$  is torsion-free because  $G$  is, and finitely presented because it is  $G$  with  $m$  relators added.  $\square$

Following Fournier-Facio [FF, §2], let  $U$  be a universal finitely presented torsion-free group, one containing a copy of every finitely presented torsion-free group [Chi]; let  $S$  be a finitely presented infinite simple torsion-free group [HL]; and let  $H_0$  be a torsion-free hyperbolic group with property (T), obtained from the density model at a parameter between  $1/3$  and  $1/2$  [KK, Ol]. Small cancellation over the relatively hyperbolic pair  $(U * H_0, U)$  embeds  $U$  in a finitely presented torsion-free quotient  $P$  of  $H_0$  [FF2, Proposition 2.3]; torsion-freeness is preserved by Osin's embedding theorem [OsSC, Theorem 2.4(5)]. Consequently,  $P$  has property (T). By universality,  $P$  contains a subgroup  $P_1 \times P_2 \times S$  with  $P_i \cong P$ . Let  $E$  be the double HNN extension of  $P$  with stable letters  $u_1, u_2$  and  $u_i P u_i^{-1} = P_i$ ; it is finitely presented and torsion-free, and its Bass-Serre action makes it acylindrically hyperbolic by the tree criterion of Minasyan-Osin [MO]. Hull's common quotient theorem [Hu, Corollary 7.4] applied to  $E$  and  $H_0$  gives a surjection  $\pi: E \rightarrow G_0$  onto a finitely presented torsion-free group with property (T), injective on a prescribed finite subset of  $E$ ; since  $E$  is finitely generated, the proof of that corollary produces  $G_0$  acylindrically hyperbolic. We run the construction with the prescribed finite subset containing an element  $s \neq 1$  of  $S$ . Then  $\pi(S) \neq 1$ , so  $\pi|_S$  is injective because  $S$  is simple.

Put  $\Gamma = \pi(P)$ ,  $t = \pi(u_1)$ , and  $J = t^{-1}\pi(S)t$ . Since  $S$  commutes with  $P_1 = u_1 P u_1^{-1}$ , the subgroup  $J$  centralizes  $\Gamma$ , and  $t\Gamma t^{-1} = \pi(P_1) \leq \Gamma$ , so  $t \in \text{Comp}_{G_0}(\Gamma)$ . Moreover  $tJt^{-1} = \pi(S) \leq \Gamma$ , and  $\Gamma$  has property (T) as a quotient of  $P$ .

**Lemma 4.3.** *Let  $\rho: G_0 \rightarrow L$  be a surjective homomorphism with  $\rho(\pi(S)) \neq 1$ . Then  $\rho(\pi(S)) \leq \mathfrak{D}_L(\rho(\Gamma))$ .*

*Proof.* Write  $\Gamma_L = \rho(\Gamma)$ ,  $t_L = \rho(t)$ , and  $J_L = \rho(J)$ . Then  $t_L \Gamma_L t_L^{-1} \leq \Gamma_L$ ,  $J_L$  centralizes  $\Gamma_L$ , and  $t_L J_L t_L^{-1} = \rho(\pi(S)) \leq \Gamma_L$ . For  $c \in J_L$  and  $\ell \in \Gamma_L$ , the commutator  $[t_L c t_L^{-1}, \ell]$  lies in  $\mathfrak{D}_L(\Gamma_L)$  by (1). As  $c$  ranges over  $J_L$ , the element  $t_L c t_L^{-1}$  ranges over  $\rho(\pi(S))$ , and  $\ell$  may be taken in  $\rho(\pi(S)) \leq \Gamma_L$ ; hence the commutator subgroup of  $\rho(\pi(S))$  lies in  $\mathfrak{D}_L(\Gamma_L)$ . Since  $\rho(\pi(S)) \neq 1$  and  $S$  is simple,  $\rho(\pi(S)) \cong S$  is perfect, so  $\rho(\pi(S)) \leq \mathfrak{D}_L(\Gamma_L)$ .  $\square$

*Proof of Theorem C.* Let  $N$  be the normal closure of  $\pi(S)$  in  $G_0$ . It is nontrivial, so Lemma 4.2 applied to  $G_0$ ,  $N$ , and  $F = \{1, \pi(s)\}$  gives  $q: G_0 \rightarrow Q$  with  $Q$  two-generated, finitely presented, torsion-free, and acylindrically hyperbolic,  $q(N) = Q$ , and  $q(\pi(s)) \neq 1$ . The group  $Q$  has property (T) as a quotient of  $G_0$ .

Let  $r: Q \rightarrow L$  be a surjection onto a nontrivial group and put  $\rho = r \circ q$ . Since  $q(N) = Q$ , the normal closure of  $\rho(\pi(S))$  in  $L$  is  $\rho(N) = L$ ; as  $L \neq 1$ , this forces  $\rho(\pi(S)) \neq 1$ . By Lemma 4.3,  $\rho(\pi(S)) \leq \mathfrak{D}_L(\rho(\Gamma))$ , and since  $\mathfrak{D}_L(\rho(\Gamma))$  is normal in  $L$  and contains a normal generating set of  $L$ , it equals  $L$ . The group  $L$  has property (T) as a quotient of  $Q$ , and  $\rho(\Gamma)$  has property (T) as a quotient of  $\Gamma$ . Theorem A, applied to  $L$  with the subgroup  $\rho(\Gamma)$  and  $K = L$ , gives  $\text{Rad}_{\text{MF}}(L) = L$ . Taking  $r$  the identity gives  $\text{Rad}_{\text{MF}}(Q) = Q$ , and a nontrivial group equal to its own MF radical is not MF.  $\square$

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